

Section 6.3

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$$1. \frac{dy}{dx} = \frac{1}{3} \left(\frac{3}{2} \right) (x^2+2)^{1/2} (2x) = x \sqrt{x^2+2} = \sqrt{x^4+2x^2}$$

$$\text{Arc length} = \int_0^3 \sqrt{1 + \left(\frac{dy}{dx} \right)^2} dx = \int_0^3 \sqrt{1 + (x^4+2x^2)} dx = \int_0^3 \sqrt{x^4+2x^2+1} dx = \int_0^3 \sqrt{x^2+1} dx = \left[\frac{x^3}{3} + x \right]_0^3 = 12$$

$$9. \frac{dy}{dx} = \frac{3}{2} \sqrt{x}$$

$$\text{Arc length} = \int_0^4 \sqrt{1 + \left(\frac{3}{2} \sqrt{x} \right)^2} dx = \int_0^4 \sqrt{1 + \frac{9x}{4}} dx = \left[\frac{8}{27} (1 + \frac{9x}{4})^{3/2} \right]_0^4 = \left(\frac{8\sqrt{10}}{27} - \frac{8}{27} \right) = \frac{8}{27} (10\sqrt{10} - 1)$$

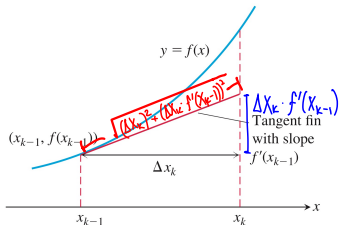
$$9. \frac{dy}{dx} = x^2 - \frac{1}{4x^2}$$

$$\therefore \text{Arc length} = \int_1^3 \sqrt{1 + \left(x^2 - \frac{1}{4x^2} \right)^2} dx = \int_1^3 \sqrt{x^4 + \frac{1}{2} + \frac{1}{16x^4}} dx = \int_1^3 \left(x^2 + \frac{1}{4x} \right) dx = \left[\frac{x^3}{3} - \frac{1}{4x} \right]_1^3 = \left(\frac{107}{12} - \frac{1}{12} \right) = \frac{53}{6}$$

$$23. \frac{dy}{dx} = \frac{1}{dx} \left(\int_0^x \cos 2t dt \right) = \cos 2x$$

$$\therefore \text{Arc length} = \int_0^{\pi/4} \sqrt{1 + (\cos 2x)^2} dx = \int_0^{\pi/4} \sqrt{1 + \cos 2x} dx = \int_0^{\pi/4} \sqrt{2} \cos x dx = \left[\sqrt{2} \sin x \right]_0^{\pi/4} = 1$$

30.



a) Length of k^{th} tangent fin over interval $[x_{k-1}, x_k]$
 $= \sqrt{(\Delta x_k)^2 + (\Delta x_k \cdot f'(x_{k-1}))^2}$ (By Pythagorean's Theorem)

b) Total length of tangent fins from a to b
 $= \sum_{k=1}^n \sqrt{(\Delta x_k)^2 + (\Delta x_k \cdot f'(x_{k-1}))^2}$

Take $\lim_{k \rightarrow \infty}$ to make it into the curve length

$$\lim_{k \rightarrow \infty} \sqrt{(\Delta x_k)^2 + (\Delta x_k \cdot f'(x_{k-1}))^2} = \int_a^b \sqrt{(dx)^2 + \left(dx \cdot \frac{dy}{dx} \right)^2}$$

$$= \int_a^b \sqrt{1 + \left(\frac{dy}{dx} \right)^2} dx = \int_a^b \sqrt{1 + (f'(x))^2} dx$$

Section 6.4

$$13. y' = \frac{x^2}{3}$$

$$\text{Area} = \int_0^2 \pi \left(\frac{x^3}{9} \right) \sqrt{1 + \left(\frac{x^2}{9} \right)^2} dx = \int_0^2 \pi \left(\frac{x^3}{9} \right) \sqrt{1 + \frac{x^4}{81}} dx$$

$$\text{Let } u = 1 + \frac{x^4}{81} \rightarrow du = \frac{4x^3}{81} dx = \frac{25}{9} \int \frac{\pi}{4} \sqrt{u} du$$

$$= \frac{\pi}{4} \left[\frac{2}{3} u^{3/2} \right]_1^{250/81} = \frac{\pi}{4} \left(\frac{250}{81} - \frac{2}{3} \right) = \frac{49\pi}{81}$$

$$18. x' = \frac{1}{2} \sqrt{y} - \frac{1}{2\sqrt{y}}$$

$$\text{Area} = \int_1^3 \pi \left(\frac{1}{3} y^{3/2} - y^{1/2} \right) \sqrt{1 + \left(\frac{1}{2} \left(\sqrt{y} - \frac{1}{\sqrt{y}} \right) \right)^2} dy = \int_1^3 \pi \left(y^{1/2} - \frac{1}{3} y^{3/2} \right) \sqrt{\frac{1}{4} \left(y + 2 + \frac{1}{y} \right)} dy$$

(for $1 \leq y \leq 3$, $\frac{1}{3} y^{3/2} - y^{1/2} < 0$)

$$= \int_1^3 \pi \left(y^{1/2} - \frac{1}{3} y^{3/2} \right) \left(\frac{1}{2} \right) \left(\sqrt{y} + \frac{1}{\sqrt{y}} \right) dy$$

$$= \frac{\pi}{2} \int_1^3 \left(-\frac{1}{3} y^2 + \frac{2}{3} y + 1 \right) dy$$

$$= \frac{\pi}{2} \left[-\frac{y^3}{9} + \frac{y^2}{3} + y \right]_1^3 = \frac{\pi}{2} \left(3 - \frac{11}{9} \right) = \frac{8\pi}{9}$$

$$29. ds = \sqrt{(dx)^2 + (dy)^2} = \sqrt{(dx)^2 + \left(\sqrt{x^4+2x^2} \cdot dx \right)^2} = \sqrt{x^4+2x^2+1} dx = (x^2+1) dx$$

$$\therefore S = \int_0^{\sqrt{2}} 2\pi x (x^2+1) dx = 2\pi \int_0^{\sqrt{2}} x^3 + x dx = 2\pi \left[\frac{x^4}{4} + \frac{x^2}{2} \right]_0^{\sqrt{2}} = 4\pi$$

$$23. ds = \sqrt{(dx)^2 + (dy)^2} = \sqrt{\left(y^3 - \frac{1}{4y^3} \right)^2 + (dy)^2} = \sqrt{(dy)^2 \left(y^6 + \frac{1}{2} + \frac{1}{16y^4} \right)} = \left(y^3 + \frac{1}{4y^3} \right) dy$$

$$\therefore S = \int_1^2 2\pi \left(\frac{y^4}{4} + \frac{1}{8y^3} \right) \left(y^3 + \frac{1}{4y^3} \right) dy = \int_1^2 \left(\frac{y^8}{4} + \frac{3}{16} y + \frac{1}{32y^3} \right) dy$$

$$= \left[\frac{y^9}{32} + \frac{3}{32} y^2 - \frac{1}{128y^2} \right]_1^2 = \left[8 + \frac{3}{8} - \frac{1}{2048} - \frac{1}{8} + \frac{1}{128} \right] = 8 \frac{527}{2048}$$

$$25. \frac{dy}{dx} = \frac{-x}{a^2-x^2} \rightarrow S = \int_{-a}^a 2\pi \sqrt{a^2-x^2} \cdot \sqrt{1 + \left(\frac{-x}{a^2-x^2} \right)^2} dx = \int_{-a}^a 2\pi \sqrt{a^2-x^2} \cdot \sqrt{\frac{a^2}{a^2-x^2}} dx$$

$$= \int_{-a}^a 2\pi a dx = 2\pi a (2a) = 4\pi a^2$$

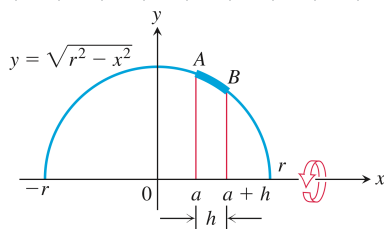
Section 6.4 (Continue)

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28.



$$\frac{dy}{dx} = -\frac{x}{\sqrt{r^2 - x^2}}$$

Area swept out by AB

$$\begin{aligned} &= \int_a^{a+h} 2\pi \sqrt{r^2 - x^2} \cdot \sqrt{1 + \left(-\frac{x}{\sqrt{r^2 - x^2}}\right)^2} dx \\ &= \int_a^{a+h} 2\pi \cdot \sqrt{r^2 - x^2} \cdot \left(\frac{r^2}{r^2 - x^2}\right) dx \\ &= \int_a^{a+h} 2\pi \cdot r dx = 2\pi r h \end{aligned}$$

$\therefore$ Area swept out by AB does not depend on the location of the interval, but it does depend on the length of the interval (h).

Section 6.5

2. a) From $F = kx$, $x = 14 \text{ cm} - 10 \text{ cm} = 4 \text{ cm}$, $F = 800$

$$800 \text{ N} = k(4 \text{ cm})$$

$$k = 200 \text{ N/cm}$$

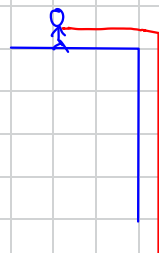
b) The spring was stretched to $12 - 10 = 2 \text{ cm}$ from the original position.

$$\text{From } W = \int F dx = \int kx dx = \int_0^2 (200 \text{ N/cm})(x) dx = [100x^2]_0^2 = 400 \text{ J}$$

c) From $F = kx$

$$1600 \text{ N} = (200 \text{ N/cm})(x) \rightarrow x = 8 \text{ cm (beyond its natural length)}$$

7.



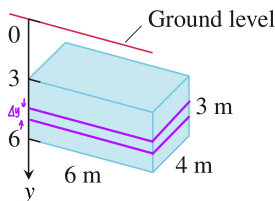
The rope's weight varies with its elevation. When the rope is $x \text{ m}$ from its original position, the remaining proportion of the rope has a weight of $0.624(50-x)$

$$\begin{aligned} \therefore W &= \int F dx \\ &= \int_0^{50} (0.624)(50-x) dx \\ &= \int_0^{50} (31.2 - 0.624x) dx \\ &= [31.2x - 0.312x^2]_0^{50} \\ &= 780 \text{ J} \end{aligned}$$

$$10. F(x) = \frac{k}{x^2}$$

$$W = \int_b^a F(x) dx = \int_b^a \frac{k}{x^2} dx = \left[-\frac{k}{x}\right]_b^a = \frac{k}{b} - \frac{k}{a}$$

14.



$$\begin{aligned} a) F &= w \cdot \Delta V = (9800 \text{ N/m}^3)(24 \Delta y \text{ m}^3) = 235200 \Delta y \text{ N, Distance} = y \\ \therefore \Delta W &= 235200 y \Delta y \end{aligned}$$

$$\text{Hence, total work required} = \int_3^6 235200 y dy = [117600 y^2]_3^6 = 3,175,200 \text{ J}$$

$$b) \text{ Time} = \frac{W}{P} = \frac{3,175,200 \text{ J}}{370 \text{ J/s}} = 8581.62 \text{ s} \approx 2 \text{ h } 23 \text{ min}$$

c) Empty tank halfway $\rightarrow$ pull water of depth $y = 4.5 \text{ m}$ to $y = 3 \text{ m}$ up

$$\text{Total work} = \int_3^{4.5} 235200 y dy = [117600 y^2]_3^{4.5} = 1,323,000 \text{ J}$$

$$\therefore \text{Time required} = \frac{1,323,000 \text{ J}}{370 \text{ J/s}} = 3575.68 \text{ s} \approx 59 \text{ min } 36 \text{ s}$$

d) In the location where water weighs $9780 \text{ N/m}^3 \rightarrow W = \int 9780 \cdot 24 y dy$

$$(a) W = 9780 \cdot 24 \cdot \left(\frac{6-3}{2}\right) = 3,168,720$$

$$(b) \text{ Time} = \frac{3,168,720}{370} = 8564.11 \text{ s} \approx 2 \text{ hr } 22 \text{ min}$$

14. d) (c) $W = 9780 \cdot 24 \cdot \left(\frac{45^2 - 3^2}{2}\right) = 1320300 \text{ J} \rightarrow \text{Time required} = \frac{1320300 \text{ J}}{370 \text{ J/s}} = 3568.38 \text{ s} \approx 59 \text{ min } 28 \text{ s}$

In the location where the water weighs $9820 \text{ N/m}^3 \rightarrow W = \int 9820 \cdot 24y \, dy$

(a) $W = 9820 \cdot 24 \cdot \left(\frac{6^2 - 3^2}{2}\right) = 3181680 \text{ J}$

(b) $\text{Time required} = \frac{3181680 \text{ J}}{370 \text{ J/s}} = 8599.13 \text{ s} \approx 2 \text{ h } 23 \text{ min}$

(c) $W = 9820 \cdot 24 \cdot \left(\frac{45^2 - 3^2}{2}\right) = 1325700 \text{ J} \rightarrow \text{Time required} = \frac{1325700 \text{ J}}{370 \text{ J/s}} = 3582.97 \text{ s} \approx 59 \text{ min } 42 \text{ s}$

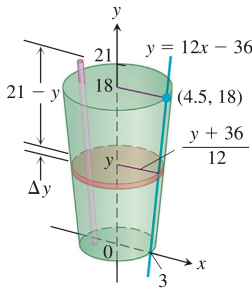
23. From $F(x) = m \left(\frac{dv}{dt}\right) = m \left(v \frac{dv}{dx}\right)$

we get $W = \int_{x_1}^{x_2} F(x) dx = \int_{x_1}^{x_2} m \left(v \frac{dv}{dx}\right) dx = \int_{v_1}^{v_2} mv \, dv \quad (v = v_1 \text{ at } x = x_1, v = v_2 \text{ at } x = x_2)$
 $= \left[\frac{1}{2}mv^2\right]_{v_1}^{v_2}$
 $= \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2$

24. $m = 60 \text{ g} = 0.06 \text{ kg}, v_2 = 50 \text{ m/s}, v_1 = 0$

$\therefore W = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2 = \frac{1}{2}(0.06 \text{ kg})(50 \text{ m/s})^2 = 75 \text{ J}$

29.



Dimensions in centimeters

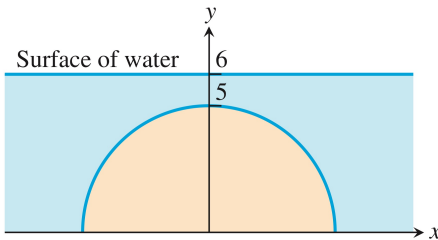
Volume of cylinder with height dy at height $y = \pi \left(\frac{y+36}{12}\right)^2 \cdot dy_{(\text{cm})}$

Weight (force) of the cylinder $= (9.8 \text{ N/kg})(0.001 \text{ kg/g})(0.8 \text{ g/cm}^3) \cdot \pi \left(\frac{y+36}{12}\right)^2 dy$
 $= 0.000171(y+36)^2 dy$

From height y , it has to be pulled up $21-y$ cm

$\therefore W = \int_0^{18} 0.000171 \cdot (y+36)^2 (21-y) dy$
 $= 0.000171 \int_0^{18} (-y^3 - 51y^2 + 216y + 27216) dy$
 $= 0.000171 \left[-\frac{y^4}{4} - 17y^3 + 108y^2 + 27216y\right]_0^{18}$
 $= 68.31 \text{ J}$

36.

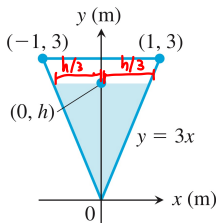


Depth = $6-y$

$L(y) = 2\sqrt{25-y^2}$

$\therefore F = \int_0^5 w(6-y) \cdot L(y) dy$
 $= \int_0^5 (9.8 \text{ N/kg})(1000 \text{ kg/m}^3)(6-y)(2\sqrt{25-y^2}) dy$
 $= 19600 \int_0^5 (6-y)\sqrt{25-y^2} dy$
 $= 19600 \left(6\left(\frac{\pi}{4} \cdot 5^2\right) + \left[\frac{1}{3}(25-y^2)^{3/2}\right]_0^5\right)$
 $= 19600 \left(\frac{75\pi}{2} - \frac{125}{3}\right) = 1492404 \text{ N}$

44.



$L(y) = 2x = \frac{2y}{3}$

$\therefore W = 25000 = \int_0^h (9.8 \text{ N/kg})(1000 \text{ kg/m}^3)(h-y)\left(\frac{2y}{3}\right) dy$

$3827 = \int_0^h (h-y)y dy = \left[\frac{hy^2}{2} - \frac{y^3}{3}\right]_0^h = \frac{h^3}{6}$

$\therefore h = 2.84 \text{ m}$

$\therefore \text{Volume of water} = \frac{1}{2}(2.84 \cdot \frac{2}{3})(2.84)(10) = 26.927 \text{ m}^3 \approx 27 \text{ m}^3$

1. One-to-one

2. Not one-to-one (It is an even function, so $f(-x) = f(x) \forall x \in \mathbb{R}$)

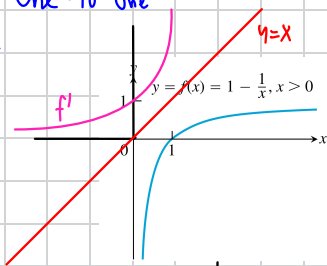
3. Not one-to-one (It is an even function, so $f(-x) = f(x) \forall x \in \mathbb{R}$)

4. Not one-to-one ($f(x) = k \quad \forall x \in [k, k+1], k \in \mathbb{Z}$)

5. One-to-one

6. One-to-one

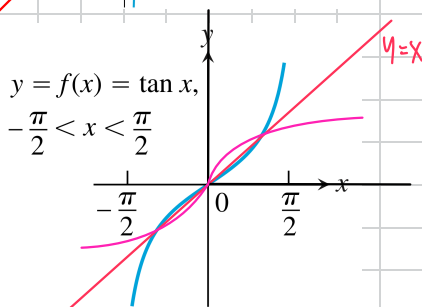
12.



$$D_{f^{-1}} = R_f = (-\infty, 1)$$

$$R_{f^{-1}} = D_f = (0, \infty)$$

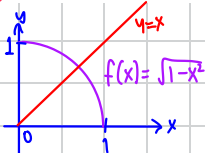
14.



$$D_{f^{-1}} = R_f = (-\infty, \infty)$$

$$R_{f^{-1}} = D_f = (-\frac{\pi}{2}, \frac{\pi}{2})$$

17. a)



The graph is symmetry about the line $y=x$

b) $f^{-1} \Rightarrow X = \sqrt{1-y^2} \quad ; 0 \leq y \leq 1$

$$X^2 = 1 - y^2$$

$$y^2 = 1 - X^2$$

$$y = \sqrt{1-X^2} \quad ; 0 \leq X \leq 1, 0 \leq y \leq 1 \quad \therefore f' = f, \text{ or } f \text{ has an own inverse.}$$

25. $f^{-1}(x) = \sqrt[5]{x} \rightarrow f(f^{-1}(x)) = (\sqrt[5]{x})^5 = x = \sqrt[5]{x^5} = f^{-1}(f(x))$, $D_{f^{-1}} = R_f = (-\infty, \infty)$, $R_{f^{-1}} = D_f = (-\infty, \infty)$

31. For $f^{-1}(x)$, we have $X = \frac{y+3}{y-2} = 1 + \frac{5}{y-2}$

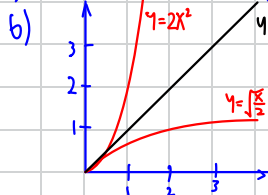
$$\therefore X-1 = \frac{5}{y-2}$$

$$y-2 = \frac{5}{X-1} ; X \neq 1$$

$$f^{-1}(x) = y = 2 + \frac{5}{X-1} = \frac{2X+3}{X-1} \rightarrow D_{f^{-1}} = \mathbb{R} - \{1\}, R_{f^{-1}} = \mathbb{R} - \{2\}$$

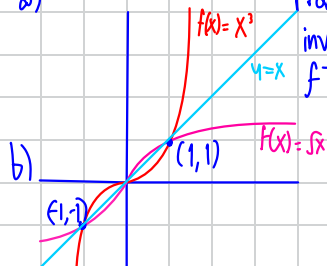
$$f(f^{-1}(x)) = \frac{\frac{2X+3}{X-1} + 3}{\frac{2X+3}{X-1} - 2} = \frac{5X}{5} = X = \frac{2(\frac{X+3}{X-2}) + 3}{\frac{X+3}{X-2} - 1} = f^{-1}(f(x))$$

38. a) $y = 2x^2 \rightarrow f^{-1}: x = 2y^2 \rightarrow y = \sqrt{\frac{x}{2}}; y \geq 0$



c) $\frac{df}{dx} \Big|_{x=5} = 4x \Big|_{x=5} = 20$
 $\frac{df^{-1}}{dx} \Big|_{x=50} = \frac{1}{2\sqrt{x}} \Big|_{x=50} = \frac{1}{20}$
 $\therefore \frac{df^{-1}}{dx} = \frac{1}{df/dx}$

39. a)



From $f(g(x)) = (\sqrt[3]{x})^3 = x = \sqrt[3]{x^3} = g(f(x))$, $f(x)$ and $g(x)$ are inverses of one another.

$f^{-1}(x): x = y^3 \rightarrow y = \sqrt[3]{x}$ (Same as $g(x)$)

c) For $f(x) = x^3$, we get $\frac{df}{dx} = 3x^2$. Hence, at $(1,1)$, the slope is $3(1)^2 = 3$ and at $(-1,-1)$, the slope is $3(-1)^2 = 3$.
 For $g(x) = \sqrt[3]{x}$, we get $\frac{dg}{dx} = \frac{1}{3\sqrt[3]{x^2}}$. Hence, at $(1,1)$, the slope is $\frac{1}{3\sqrt[3]{1^2}} = \frac{1}{3}$ and at $(-1,-1)$, the slope is $\frac{1}{3\sqrt[3]{(-1)^2}} = \frac{1}{3}$.

d) At $x=0$, the slope of $f(x)$ is $\frac{df}{dx} \Big|_{x=0} = 3(0)^2 = 0 \therefore$ The line is $y-0=0 \rightarrow y=0$ and the slope of $g(x)$ is $\frac{dg}{dx} \Big|_{x=0} = \frac{1}{3\sqrt[3]{0^2}}$ which is undefined, hence the line is vertical as $x=0$.

44. $\frac{dg^{-1}}{dx} \Big|_{x=0} = \frac{1}{\frac{dg}{dx} \Big|_{x=f^{-1}(0)=0}} = \frac{1}{2} \quad (f(0)=0 \text{ as graph passes origin})$

56. Let $a, b \in D_h$ such that $h(a) = h(b)$
 $\therefore \frac{1}{f(a)} = \frac{1}{f(b)}$

Multiply both sides by $f(a)f(b) \neq 0$: $f(b) = f(a) \rightarrow a=b \quad (f \text{ is 1-1})$

Hence, $h(x)$ is also one-to-one.

59. As $g(f(x)) = x$
 $(g(f(x)))' = x'$
 $g'(f(x)) \cdot f'(x) = 1$